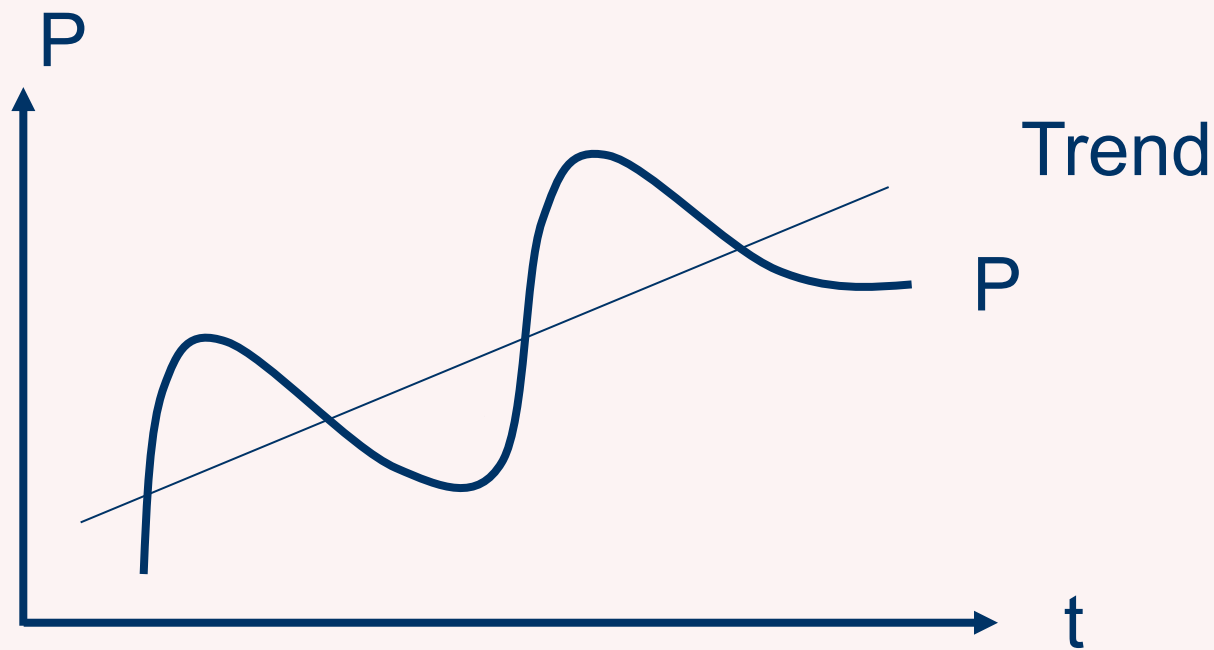


GY457
Applied Urban and Real Estate Economics

Block II—Topic 4:

Real Estate Cycles: Some Stylized Facts & Theory



Last lecture and seminar

- Looked at demand and supply curves and how supply and demand elasticities can be identified/estimated
- One finding: More developed places have more inelastic supply & prices respond more strongly to demand shocks
 - ▶ Implies greater price volatility & cyclical

This lecture & next two lectures

- **Look at phenomenon of ‘real estate cycles’ in more depth**
- **What is a cycle?**
 - ▶ Not same as volatility
 - ▶ A series of events that are regularly repeated in same order
 - ▶ In technical terms: Price changes are characterized by **serial correlation** and **mean reversion**
- **What determines real estates cycles?**
 - ▶ Look at theories and supporting evidence...

Overview 'Real Estate Cycle' Lectures: Topics 4-6

1. Real estate cycles: some stylized facts
 2. Theoretical considerations ("Econ 101") & initial evidence
 3. Real estate cycles driven by exogenous demand shocks
 - The role of long-term supply constraints and demand shocks: The case of England
 - The role of credit conditions
 4. Some further stylized facts and puzzles
 5. Endogenously driven cycles and behavioural explanations: Theories and evidence
 6. Price vs. rent dynamics & commercial vs. residential RE
 7. Conclusions and policy implications
- Topic 4
(Handout 4)
- Topic 5
(Handout 5)
- Topic 6
(Handout 6)

Overview 'Real Estate Cycle' Lectures: Topics 4-6

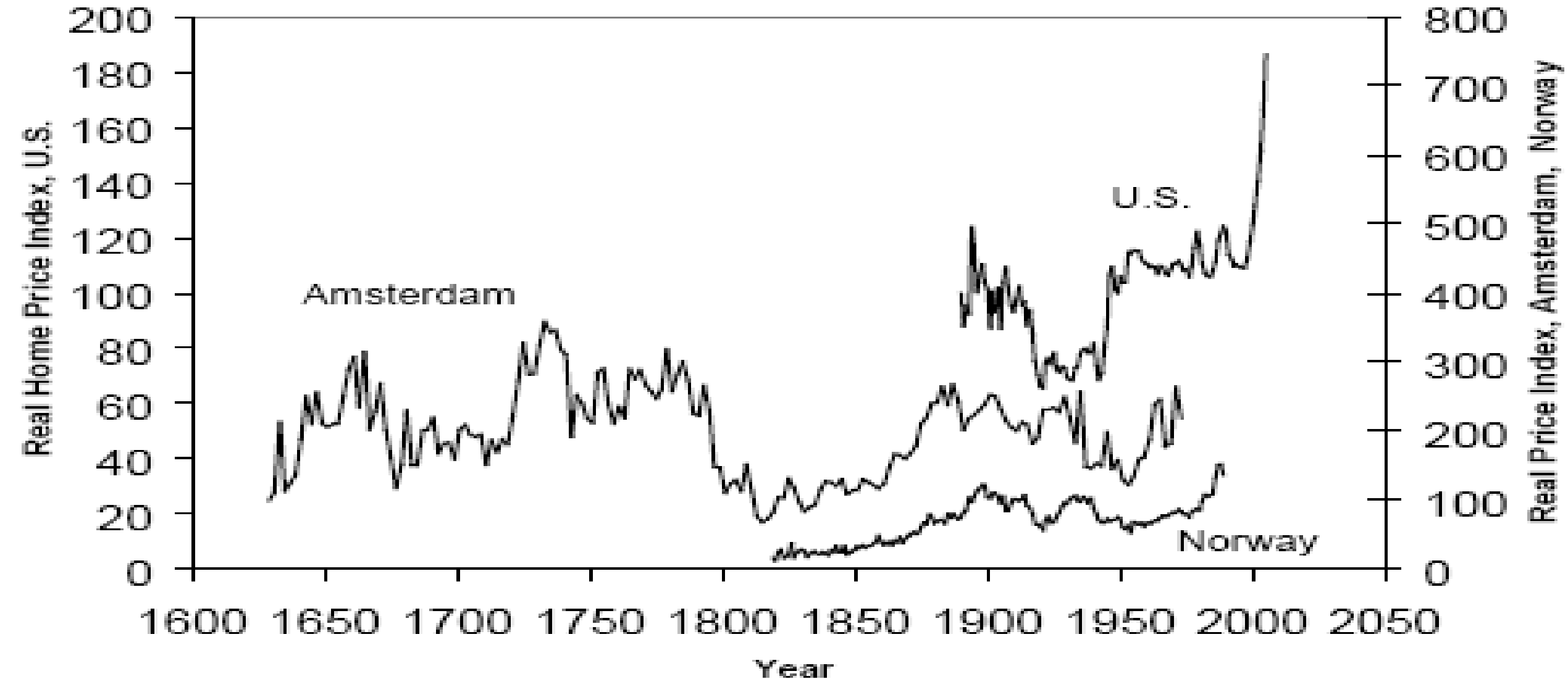
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Overview

Stylized Fact 1

Real estate has ‘always’
been subject to strong
price volatility.

Historical real house price indices in Amsterdam, Norway and the U.S.

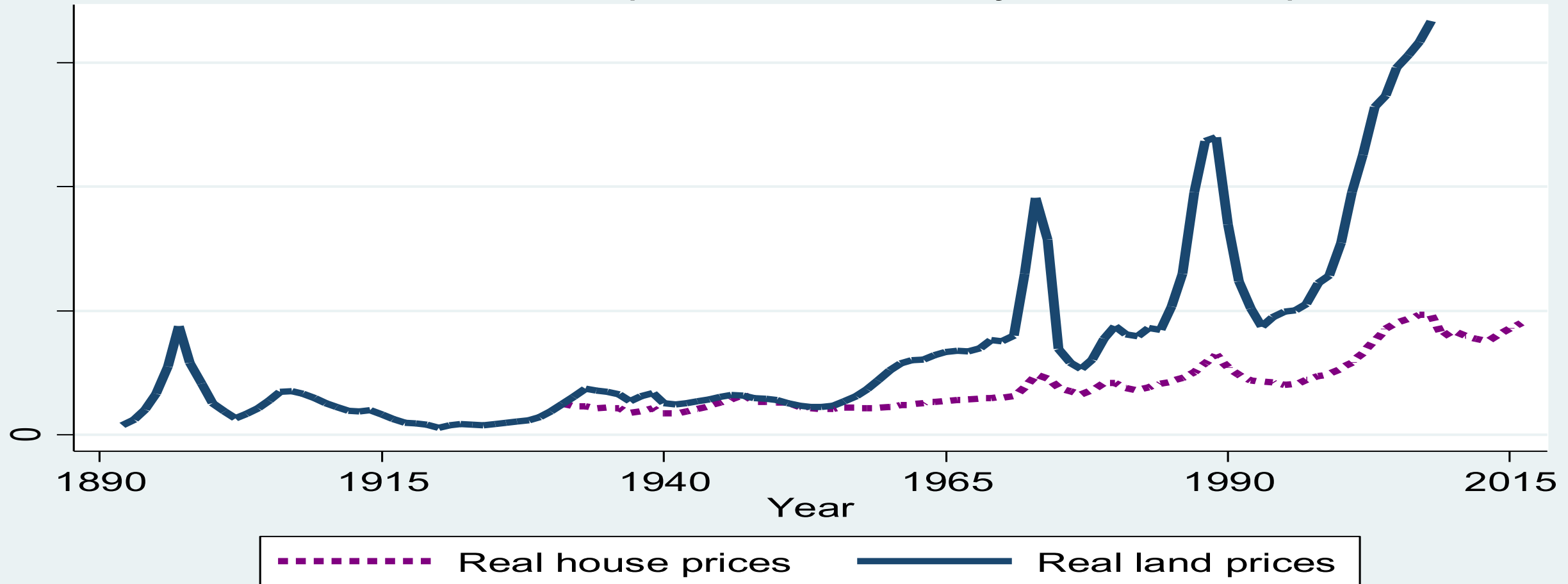


Source: Shiller (2006) based on Shiller (2005), Eichholtz (1997), Eitrheim and Erlandsen (2004)

Volatility has seemingly increased in recent decades (...at least in UK)

Real land and house price indices (1931=100)

Note: House and land price data for war years are interpolated



Source: Cheshire (2009) and own calculations for 2008 onwards / Land Registry & Nationwide

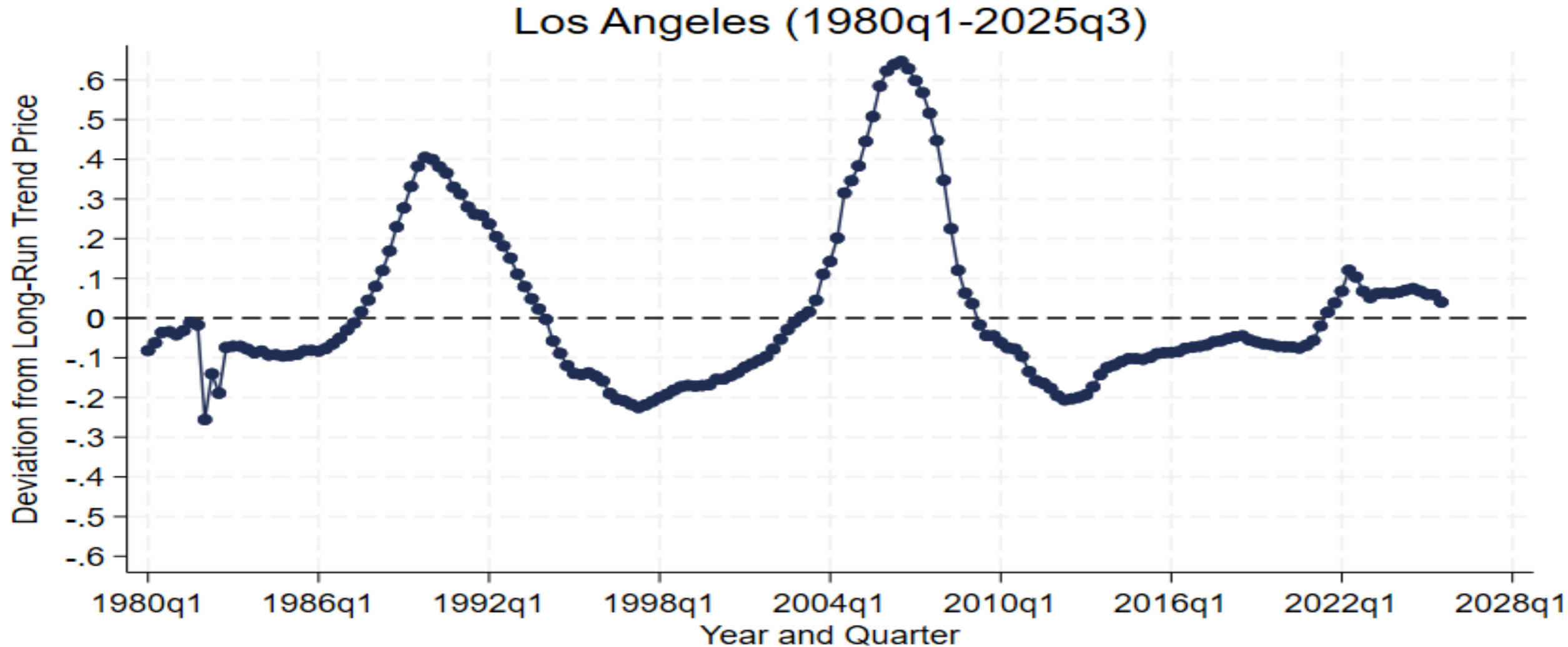
Overview

Stylized Fact 2

Various key measures of
residential and commercial
property markets behave
cyclically.

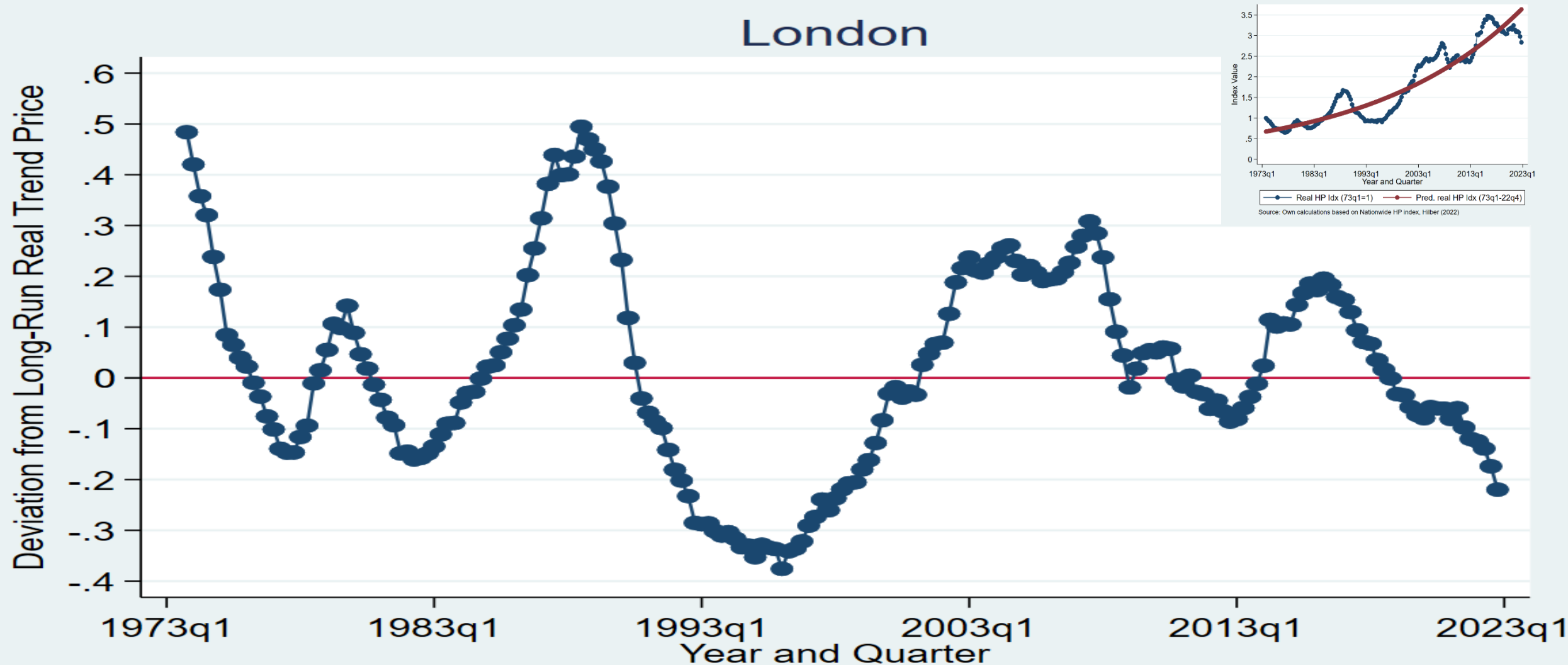
*(i.e., measures are serially correlated
and mean-reverting)*

Example: Deviation of house prices from long-run trend in LA

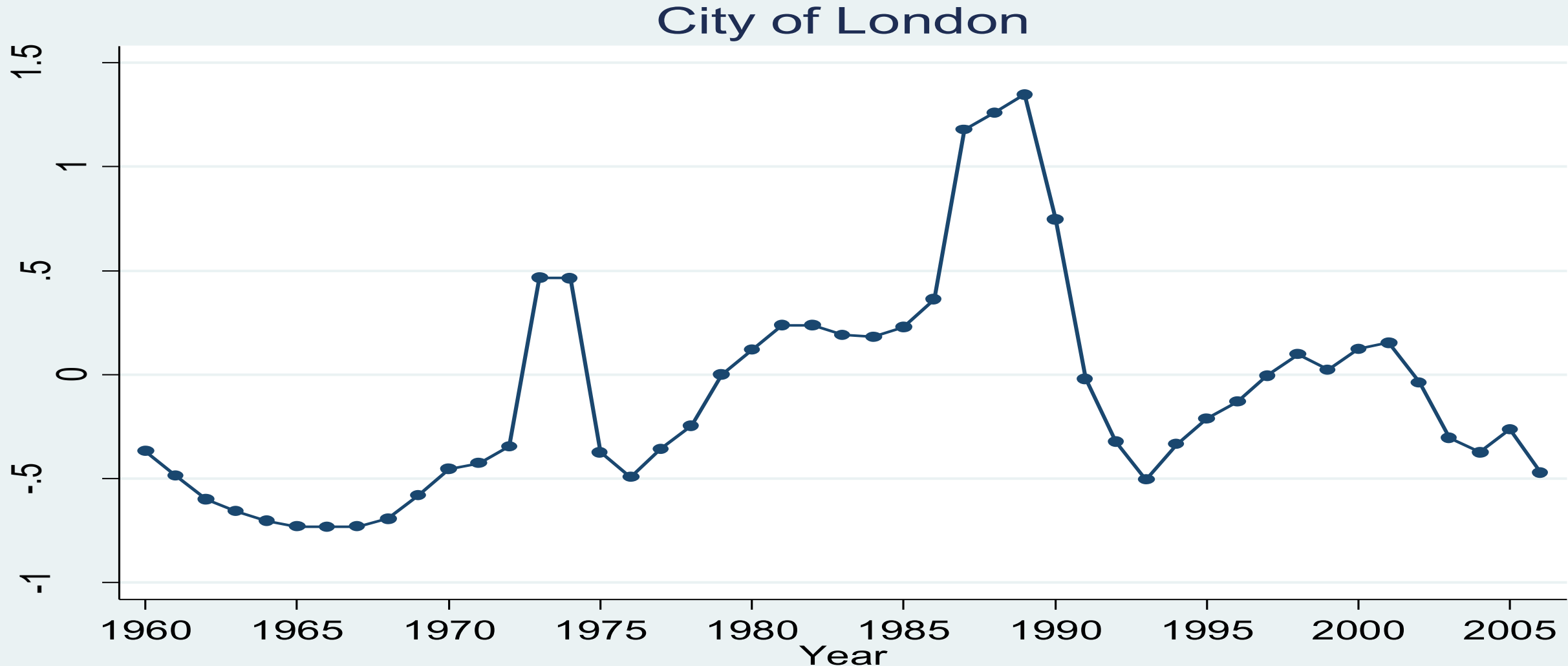


Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Example: Deviation of **real house prices** from long-run trend in London (1973q4-2022q4)



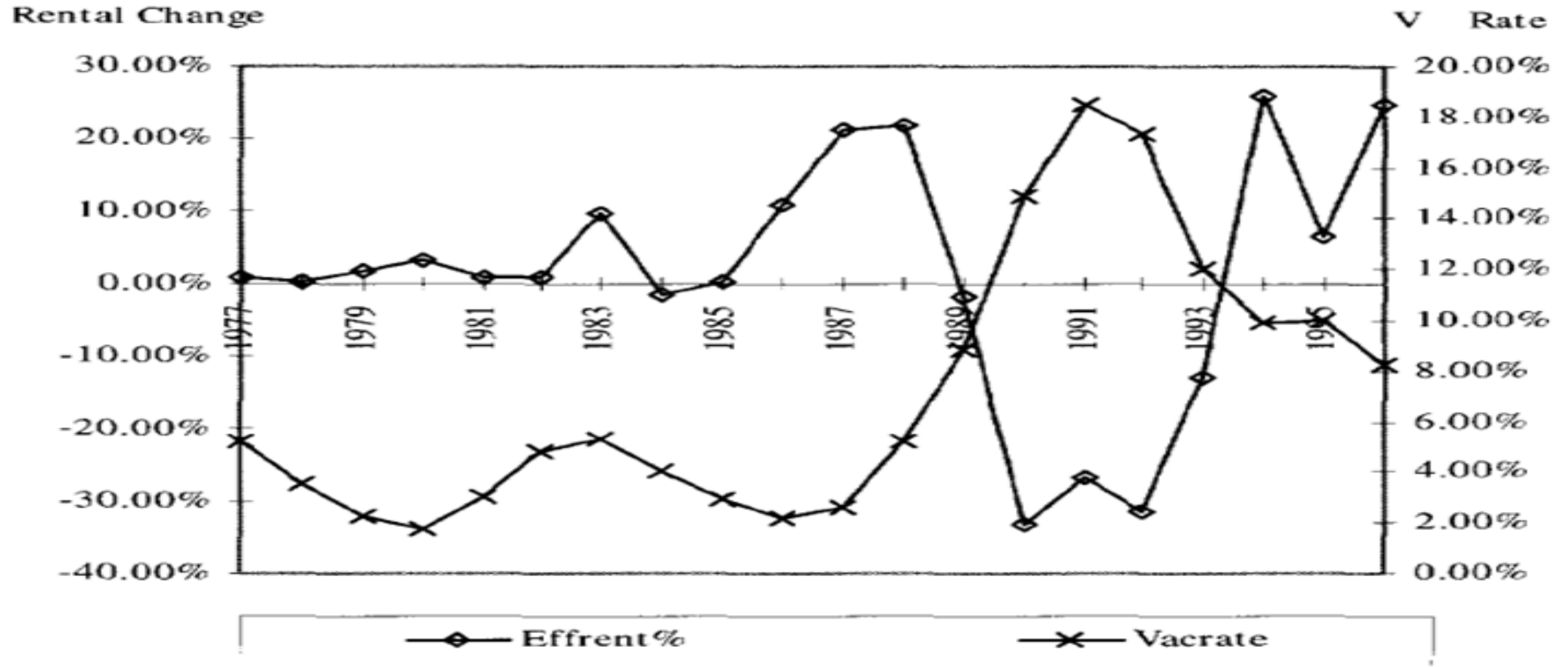
Example: Deviation of office prices from long-run trend in City of London (1960-2006)



Source: CBRE & Own Calculations, Hilber (2006)

Example: London office vacancy rates and effective rents

Figure 1 ■ Change in real effective rent and the vacancy rate.



Source: Hendershott *et al.* (1999)

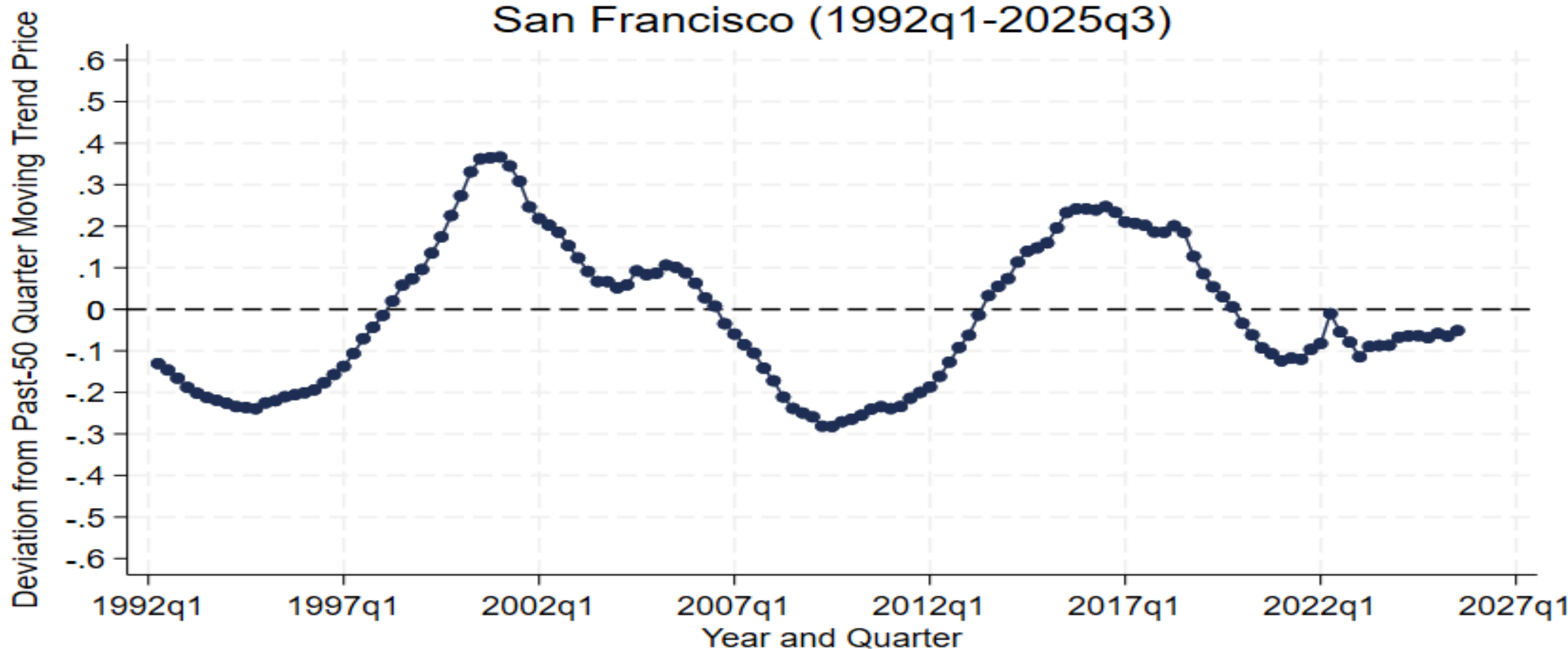
Overview

Stylized Fact 3

The magnitude, timing and duration of property cycles varies substantially **across markets** and **property types**.

Example: Housing market of SF (CA)

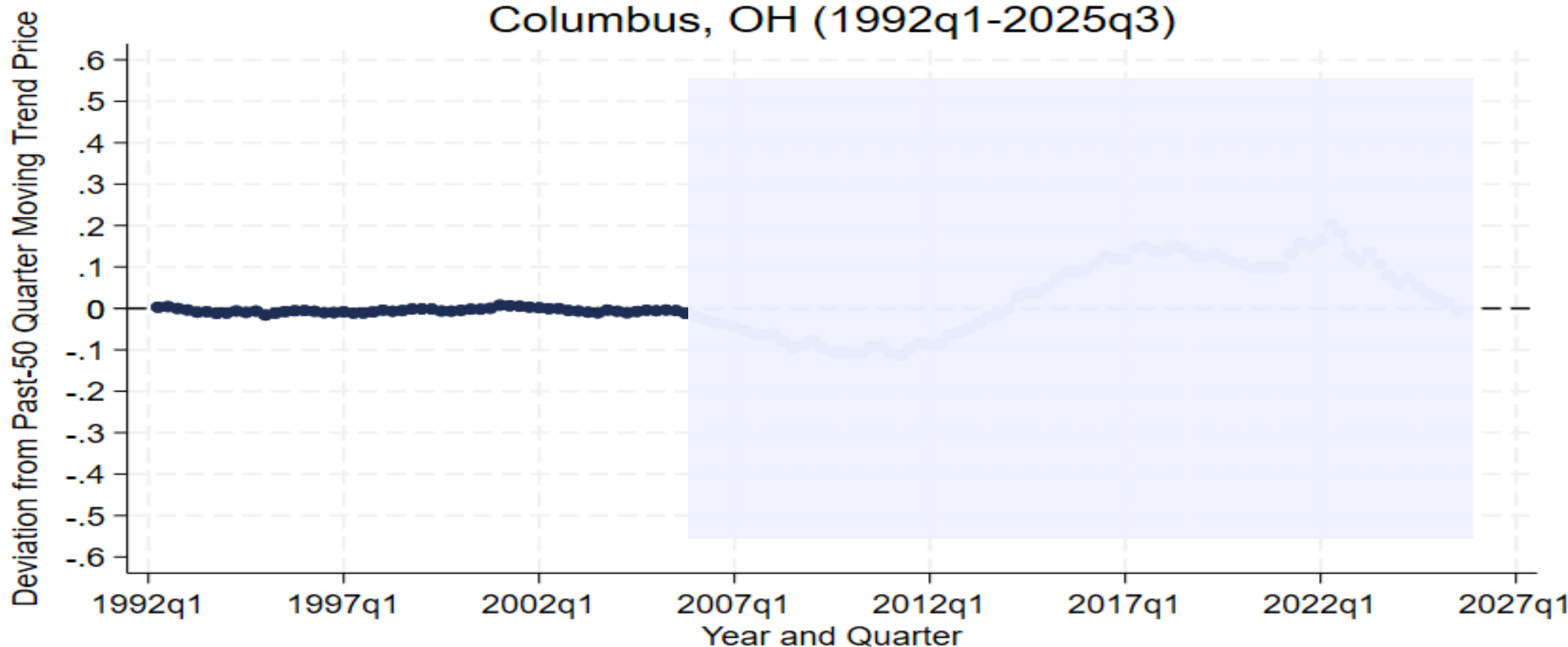
- Deviation from 50q moving trend price



Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

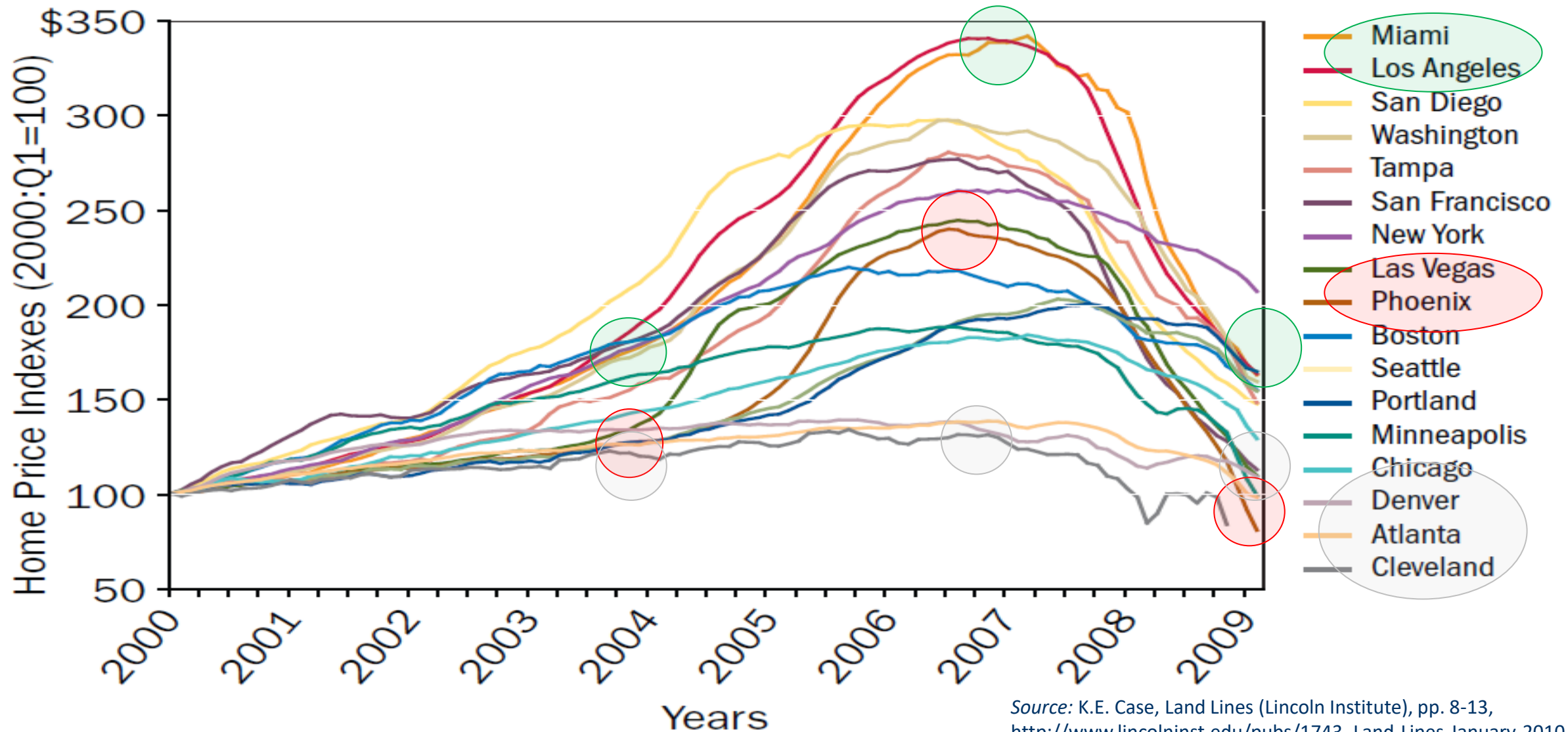
Example: Housing market of Columbus (OH)

– Deviation from 50q moving trend price

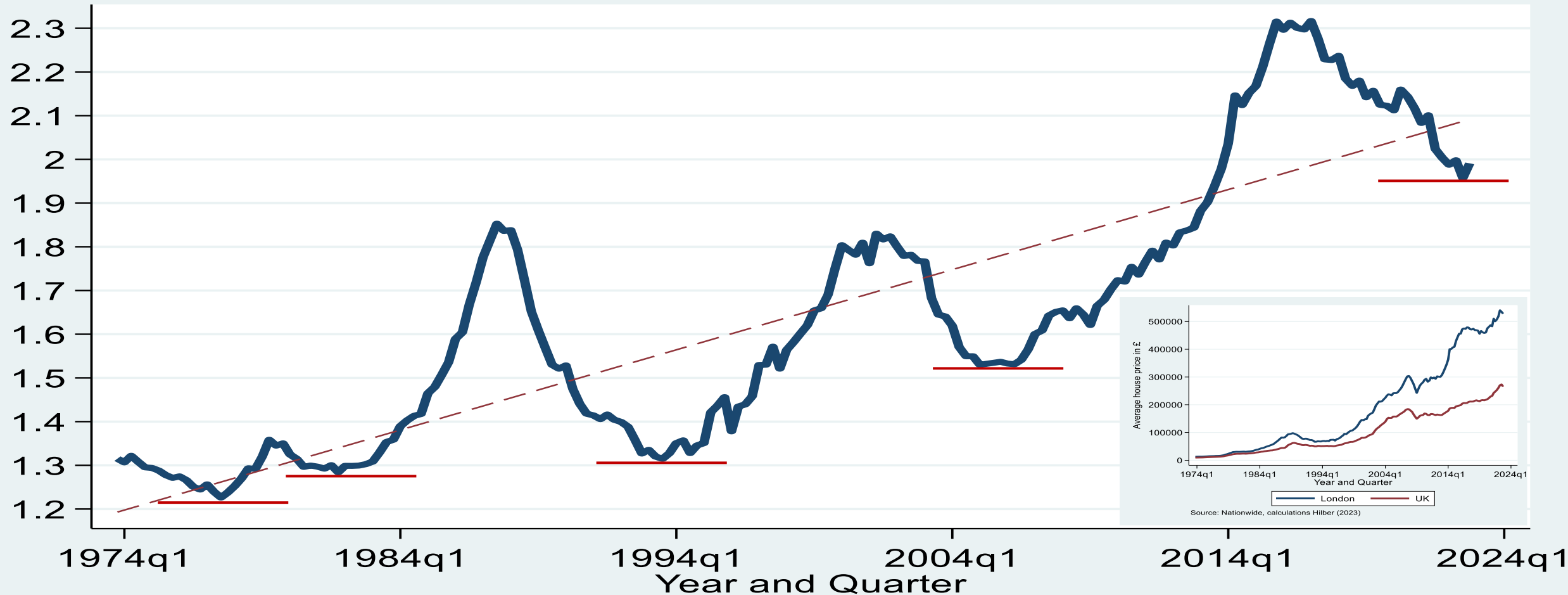


Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Housing transaction prices in 17 MSAs



House prices in London increase more strongly and are more volatile (1973q4-2022q4)



— Av. house price in London rel. to UK (ratio) - - - Fitted values

Source: Nationwide, calculations Hilber (2023)

Example: Cycle duration

Duration of full HP cycle based on most recent full & clearly defined cycle

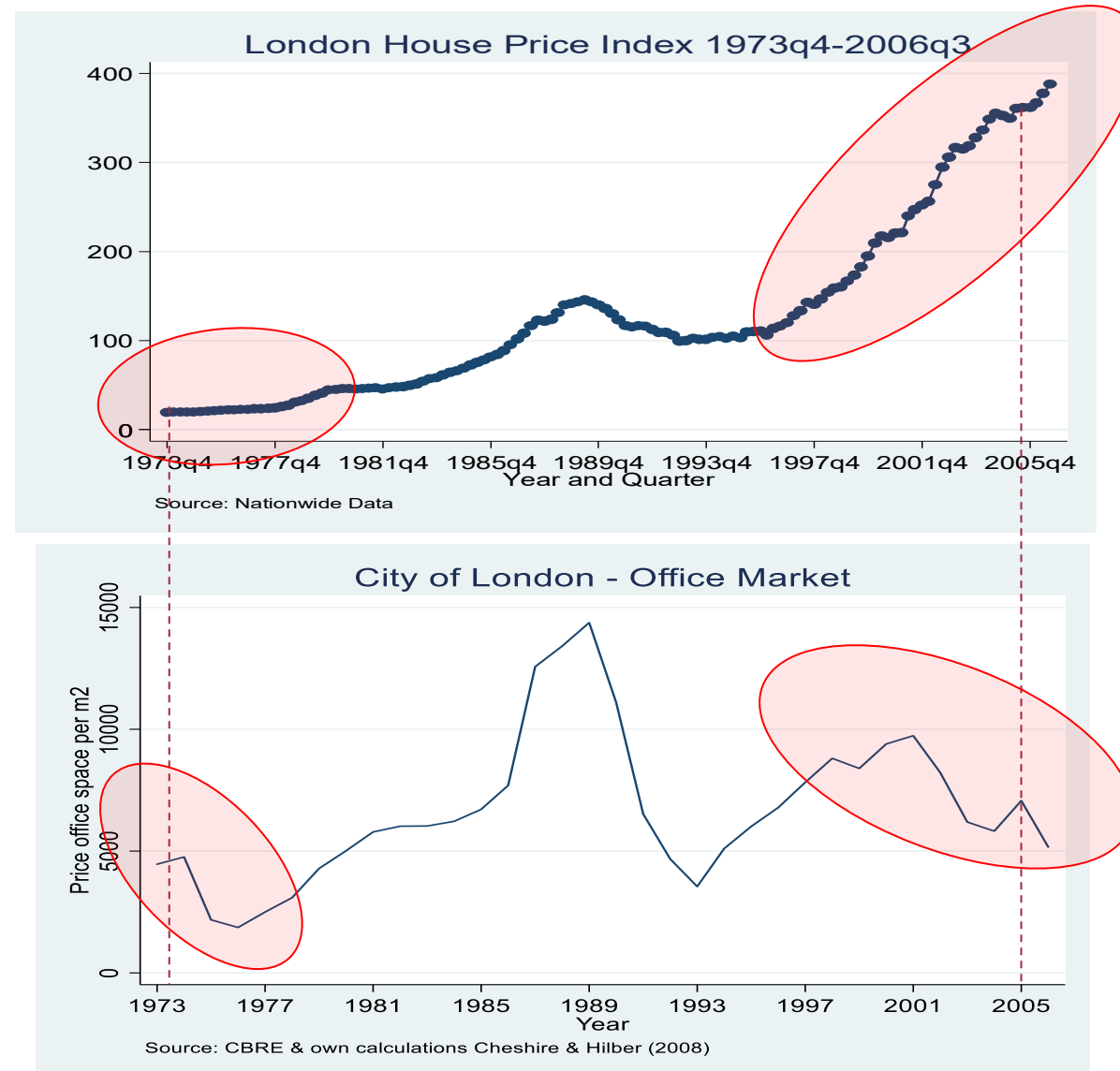
MSA	Start Date of 1 st Boom/Bust Cycle	Start Date of 2 nd Boom/Bust Cycle	Duration in Years
Phoenix	1980	1998	18
Fort Worth	1982	1999	17
Dallas	1982	1999	17
...	...	...	...
San Diego	1988	2000	12
Grand Rapids	1988	1999	11

Source: Hilber (2003), own calculations
based on OFHEO data, N = 39

Cycle duration (in quarters) across 19 OECD countries (Bracke 2013, JHE)

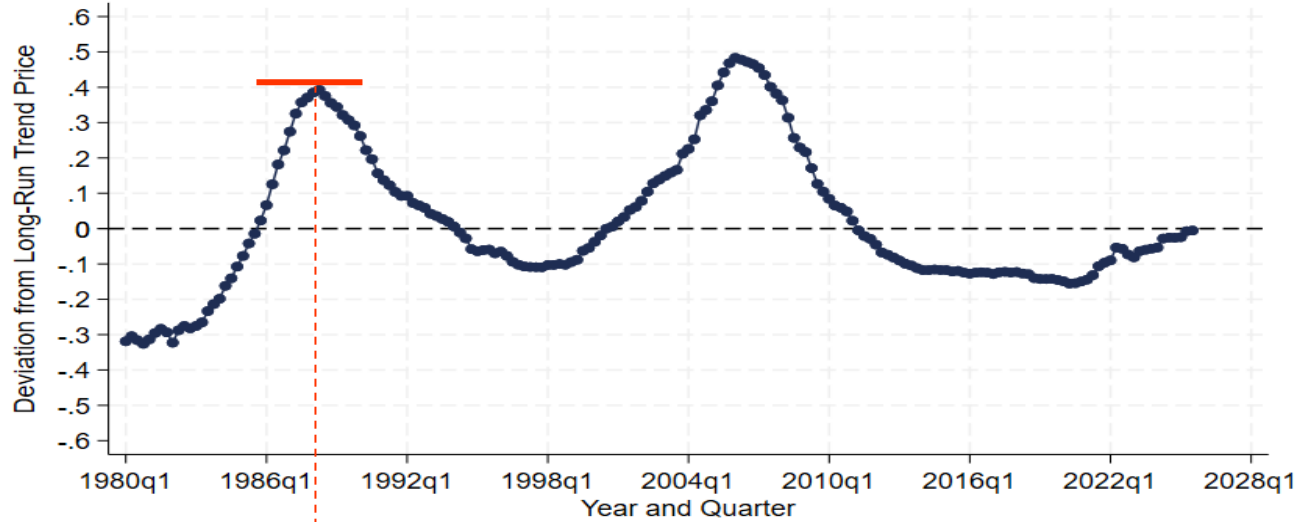
	Pct10	Pct25	Median	Pct75	Pct90
Completed upturns	8	12	21	32	47
Completed downturns	7	13	17	23	32

Residential vs. office market in London



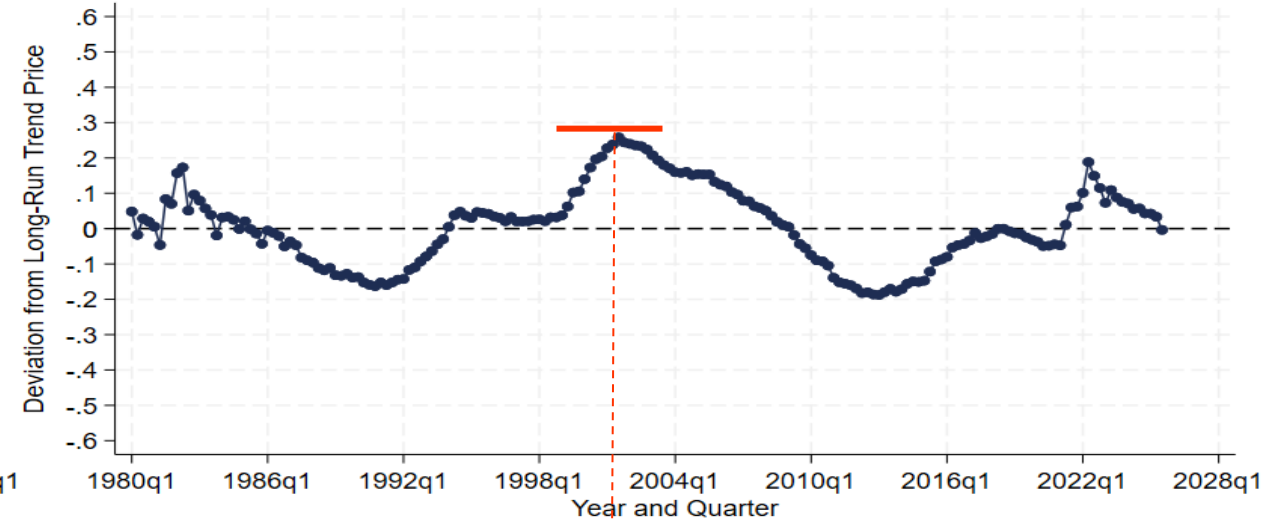
Housing **cycles** are **local** in nature

New York (1980q1-2025q3)



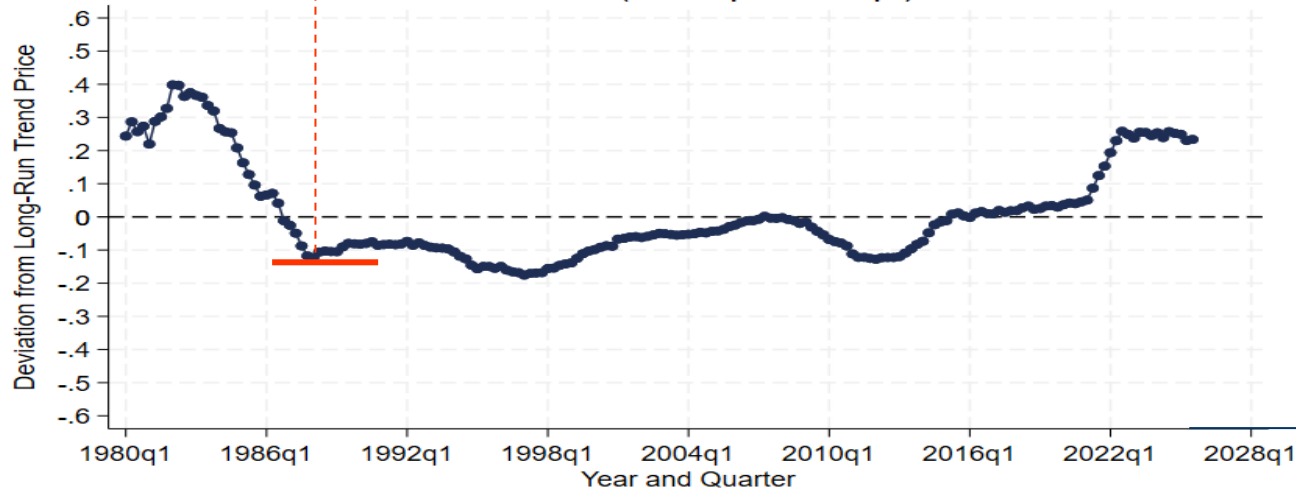
Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Boulder (1980q1-2025q3)



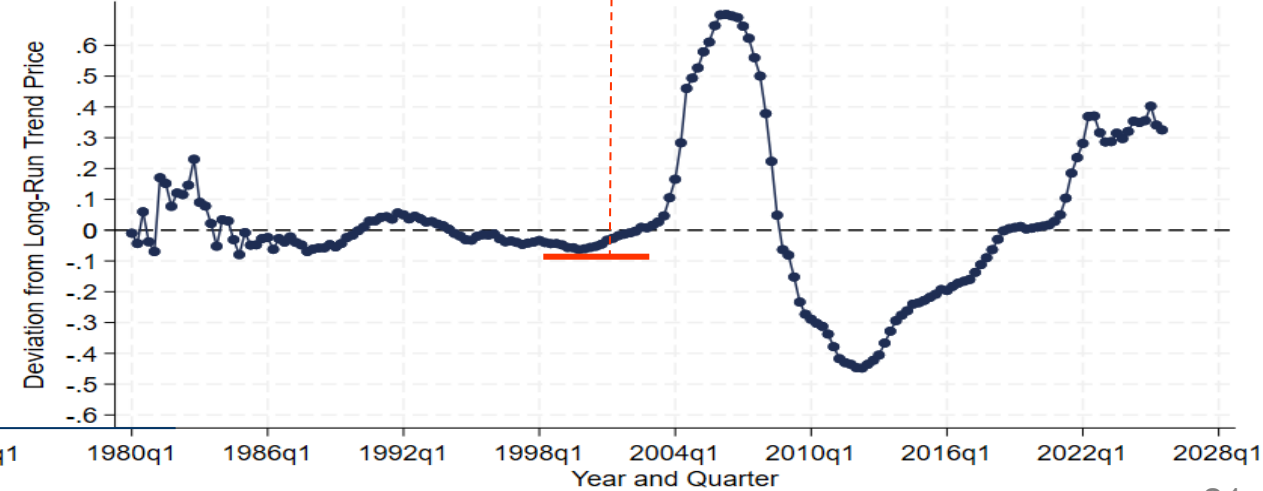
Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Houston (1980q1-2025q3)



Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Las Vegas (1980q1-2025q3)



Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

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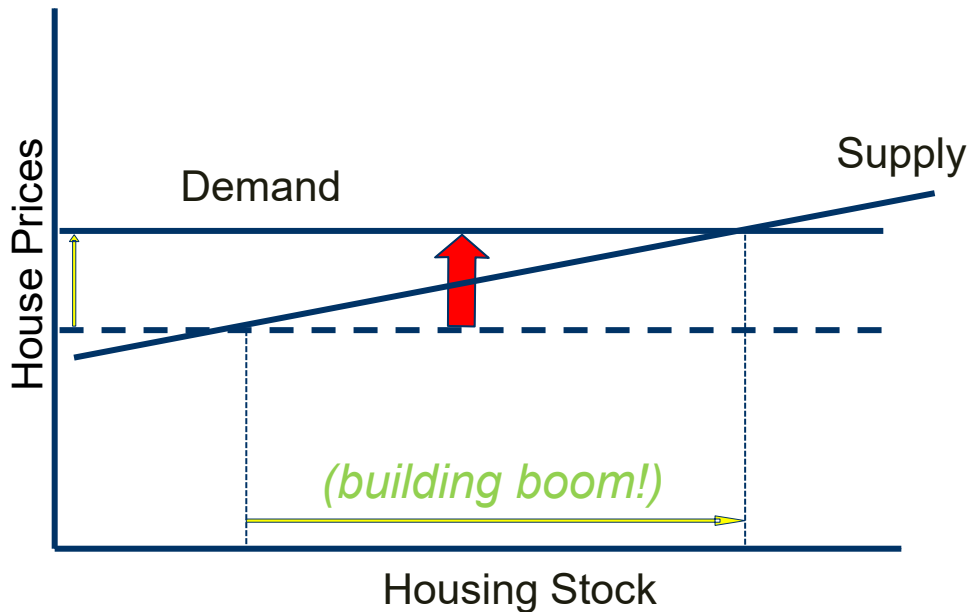
Some theoretical considerations...

- Consider a metro area with a large number of small local jurisdictions $j=1, \dots, J$
 - ▶ All jurisdictions are perfect substitutes (same amenities, same LPGs and taxes)
 - ▶ Households have identical preferences (same WTP)
 - ▶ Households relocate without cost (no attachment)
- **Question:** Should the availability of land in jurisdiction j matter for capitalization of demand shocks? **(DISCUSS)**

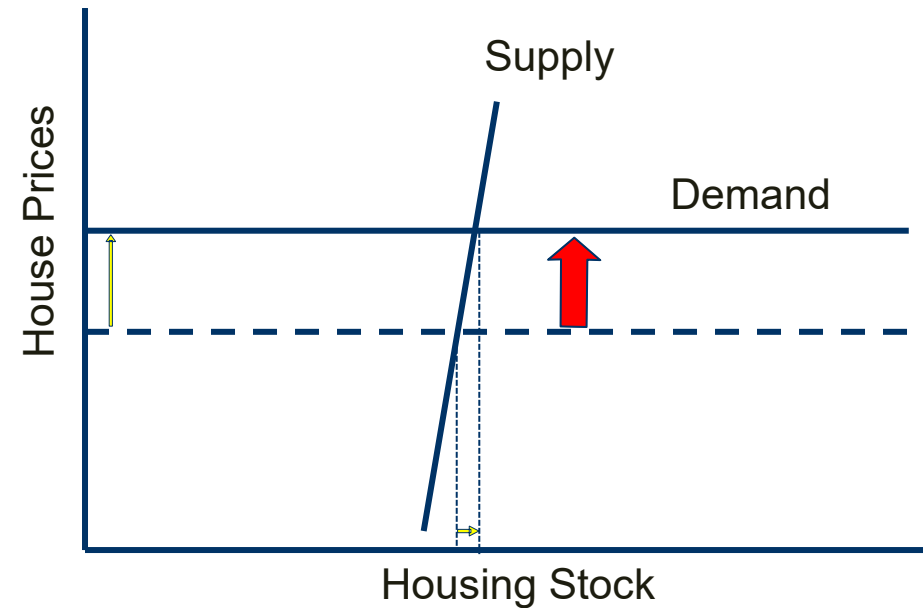
Intuitive explanation

The case of perfectly elastic demand

Unconstrained location (small town A)



Constrained location (neighboring small town B)



- ⇒ Slope of supply curve does not affect extent of price capitalization (always 100%) but matters greatly for new construction!
- ⇒ Hedonic model assumes perfectly elastic demand!

Glaeser and Ward (2009, JUE)

“There are so many **close substitutes** for most towns [in Greater Boston] that we would not expect restricting of housing supply in one town to raise prices in that town relative to another town with similar demographics and density levels. Restrictions on building in one suburban community should not raise prices in that community relative to another town **with equivalent amenities**, any more than restrictions on the production of Saudi Arabian crude will raise the price of Saudi Arabian crude relative to Venezuelan crude. Of course, Saudi Arabia’s quantity restrictions will still raise the global price of oil, but this cannot be seen by comparisons of prices across oil producers.”

How realistic is this case?

- **Two neighboring small towns**
 - ▶ Often *close substitutes* but not always (school quality often very different; each town has *some* unique features; attachment to place of birth)
- **City centre vs. small town at edge**
 - ▶ *Poor substitutes*: very different amenities, local public services, commuting times
- **Two metro areas in same country**
 - ▶ *Very poor substitutes*: NYC very different from Columbus (Ohio)

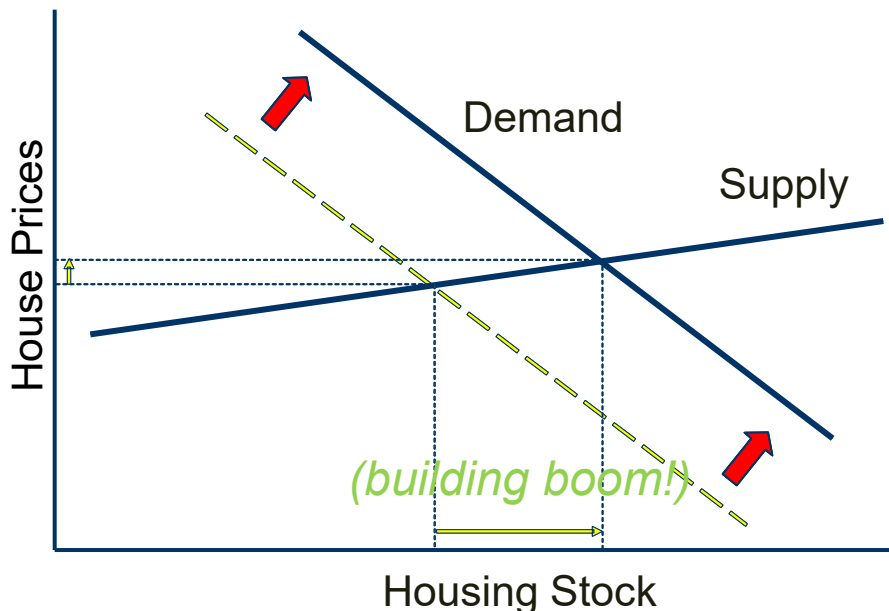
How realistic is this case? (Cont.)

- In world with imperfect substitutability of land
 - ▶ Relocation costs matter
 - ▶ Preferences (tastes) matter (e.g. attachment to place of birth; love for mountains/solitude)
- **Why?**
 - ▶ Take grant example: If relocation costs are $> 1000\text{£}$, no household will move (unless HHs experience ‘mobility shock’)
 - ▶ More generally: heterogeneous preferences & imperfect substitutability make **local demand curves downward sloping** (because each HH has different WTP for attributes of location!)

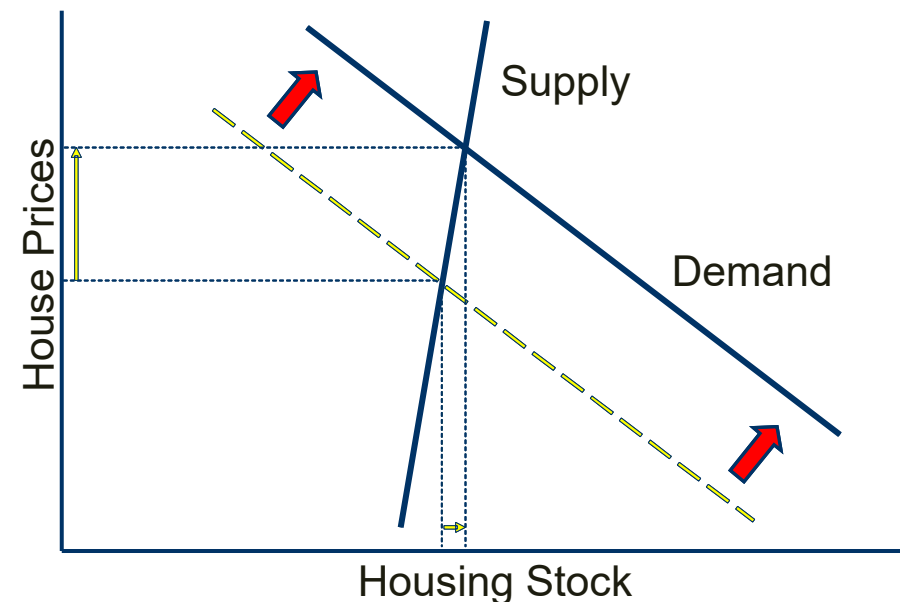
The role of supply constraints

- Consider labour demand shock in two locations that are not perfect substitutes & HH differ in tastes for amenities & local public services...

Unconstrained location (Columbus, OH)



Constrained location (SF, LA, London)



Just in Theory?

- Look at
 - ▶ **city that has very little undeveloped land & is tightly regulated** and compare with
 - ▶ **city with plenty of open land** in the surrounding area & **few land use restrictions**

A prime example of a city with inelastic land supply

San Francisco (CA)

Little undeveloped land +
geographical constraints +
tight land use control



A prime example of a city with plenty of open land

Columbus (OH)

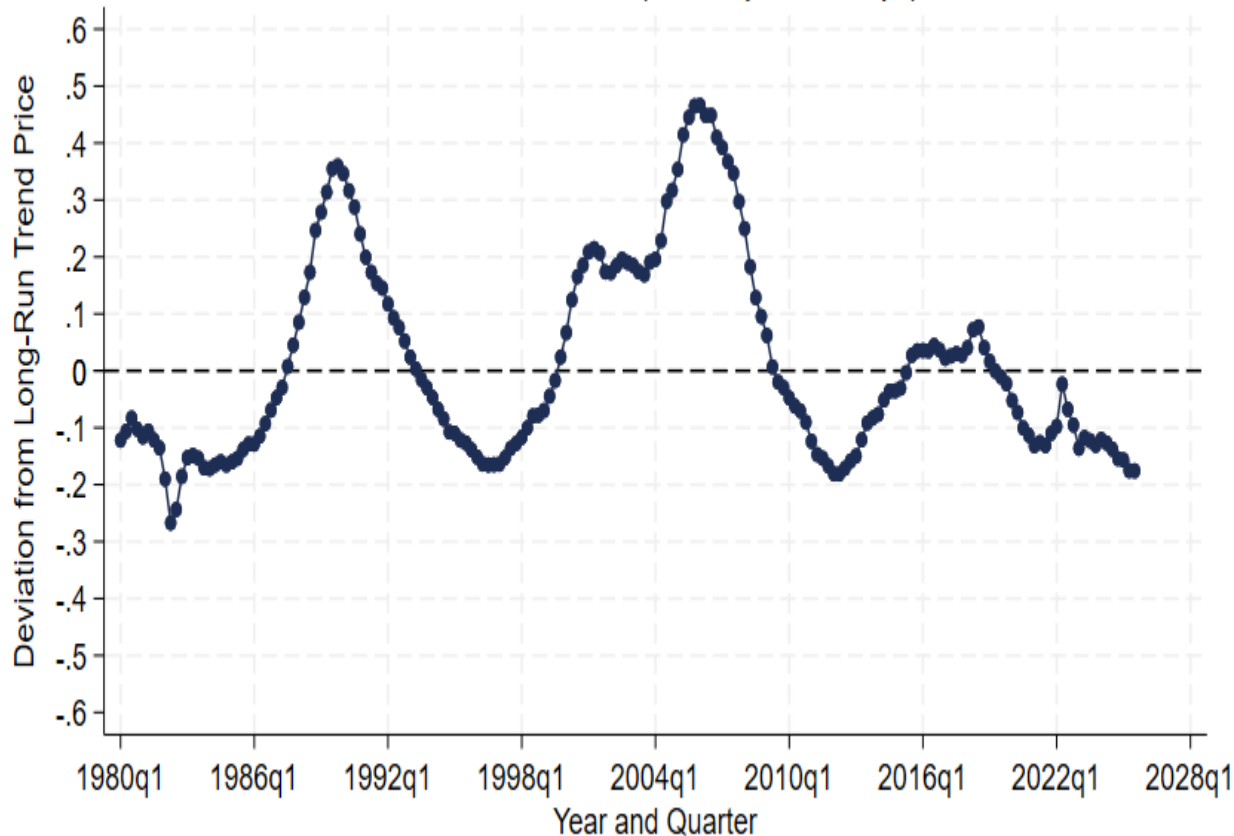
Plenty of open land surrounding city +
no geographical constraints +
lax land use controls



San Francisco (CA) vs. Columbus (OH)

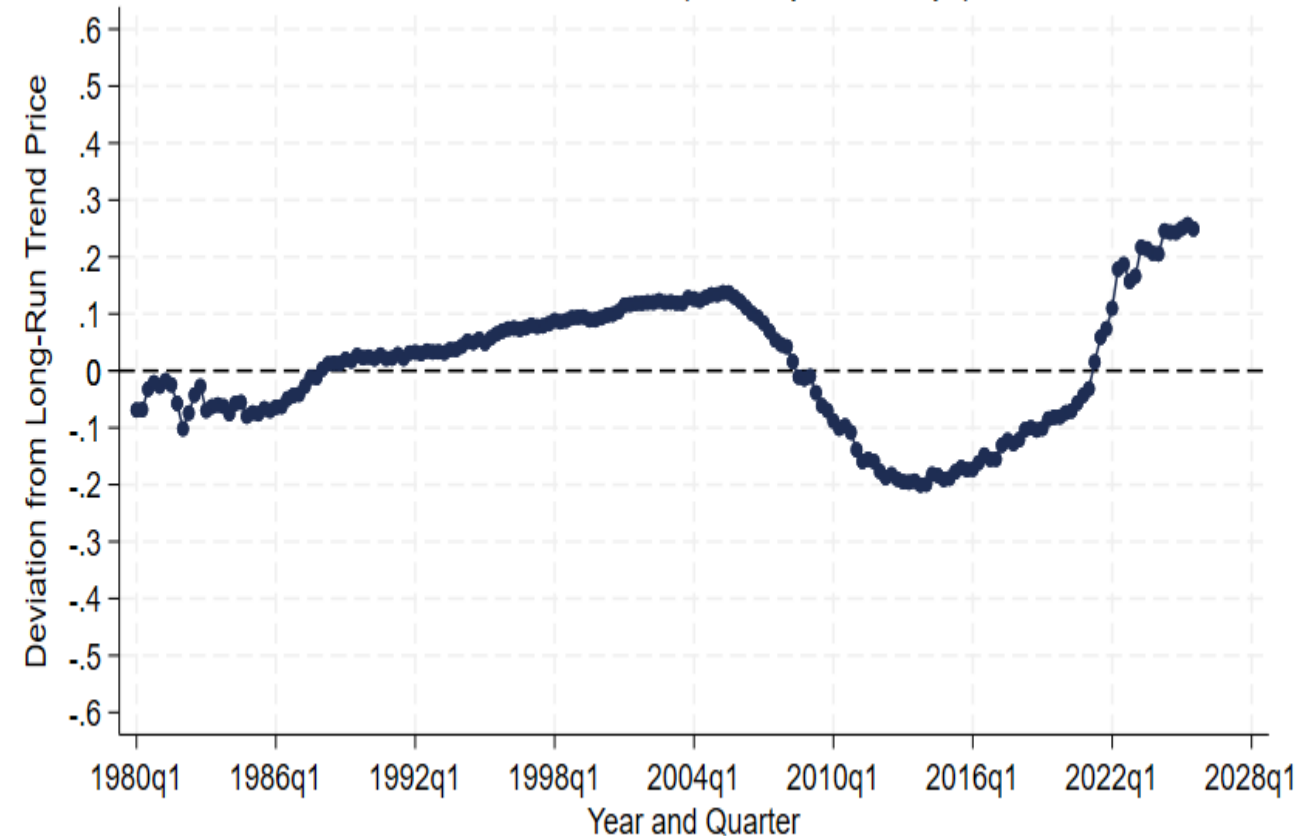
Deviation of house price index from long-run trend

San Francisco (1980q1-2025q3)



Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

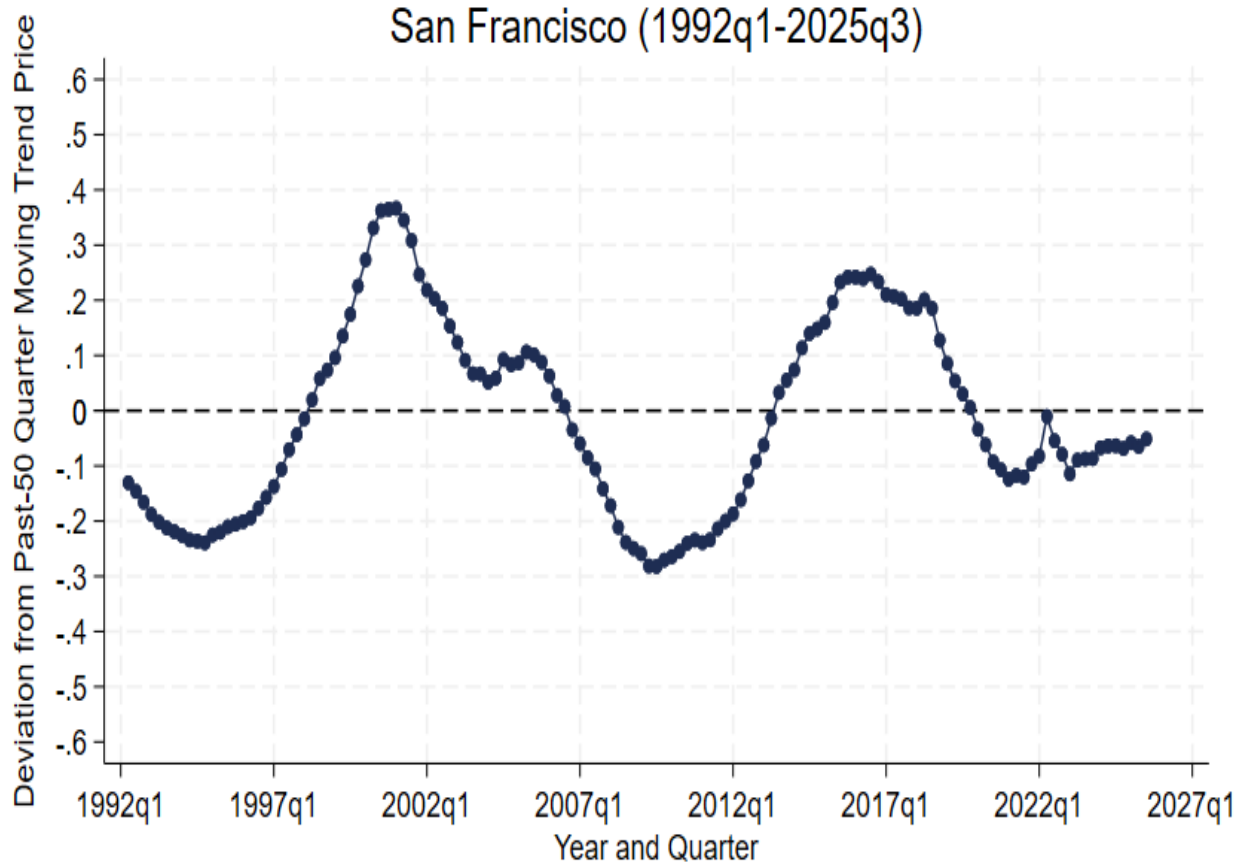
Columbus, OH (1980q1-2025q3)



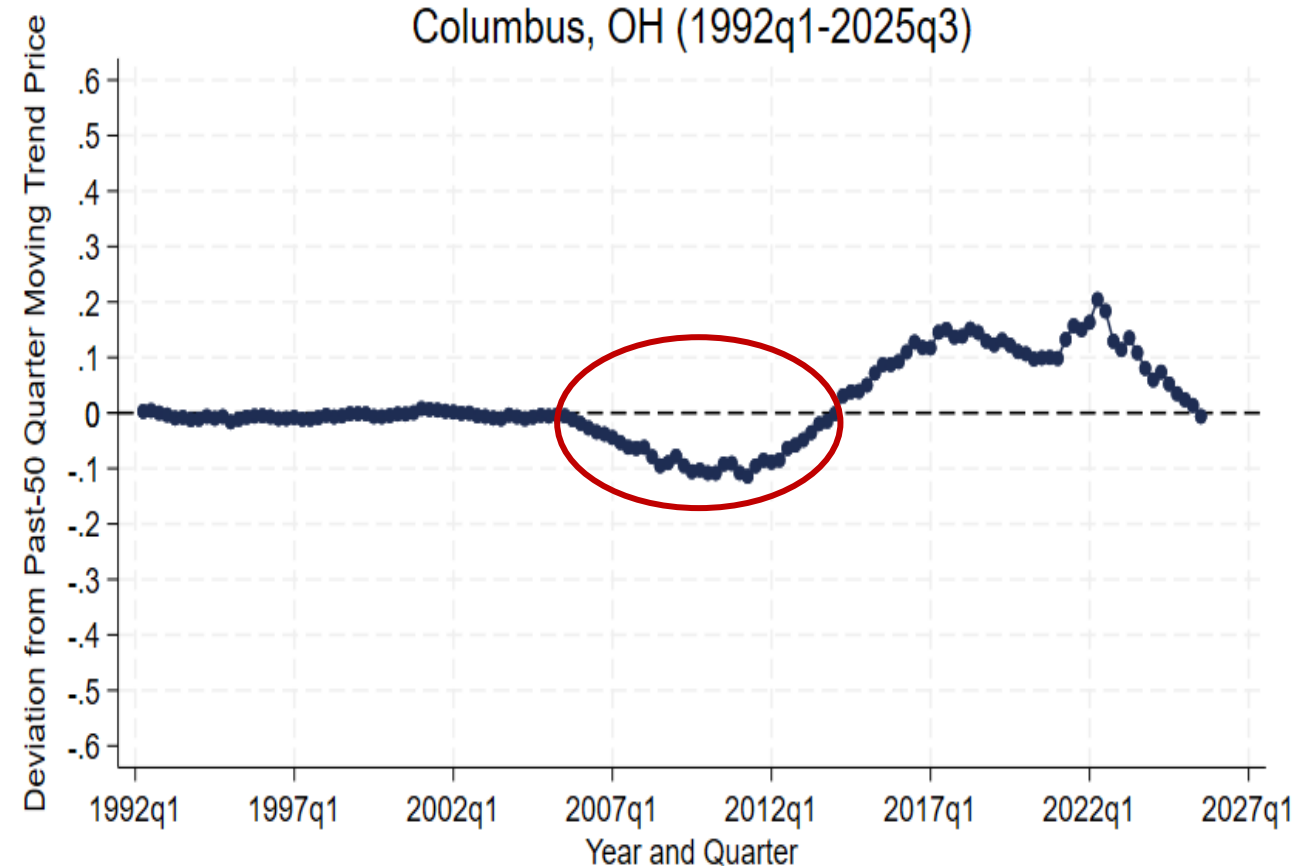
Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

San Francisco (CA) vs. Columbus (OH)

Deviation of house price index from moving average
(last 50 quarters)



Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

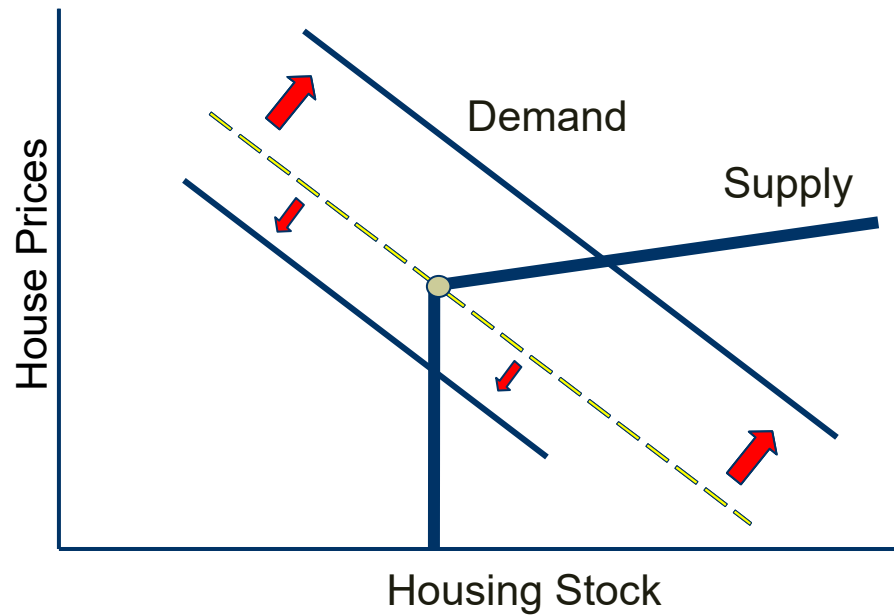


Source: Own calculations based on FHFA all-transactions HP index, Hilber (1/2/2026)

Excursus: The housing supply curve is 'kinked downwards'

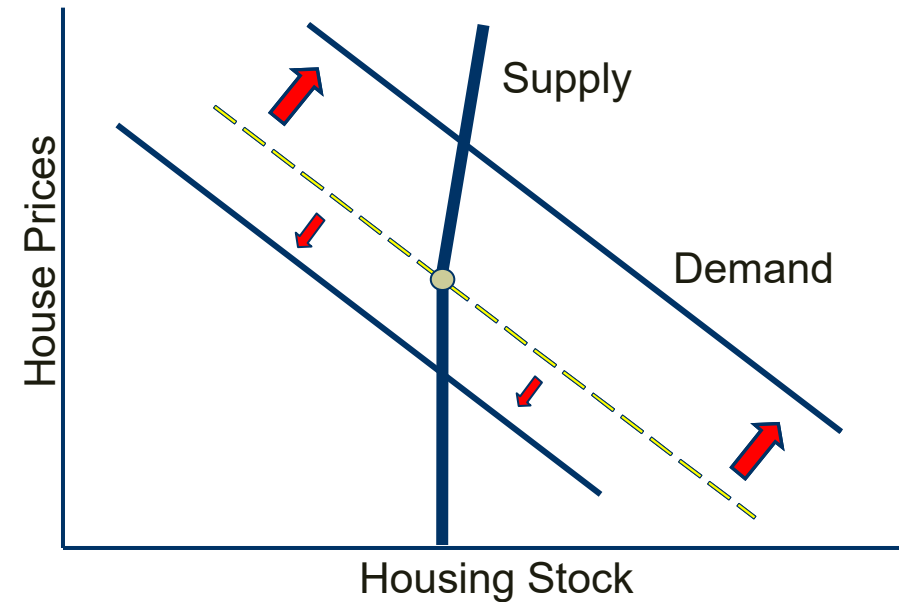
Unconstrained location

Columbus (OH), Chattanooga (TN-GA)



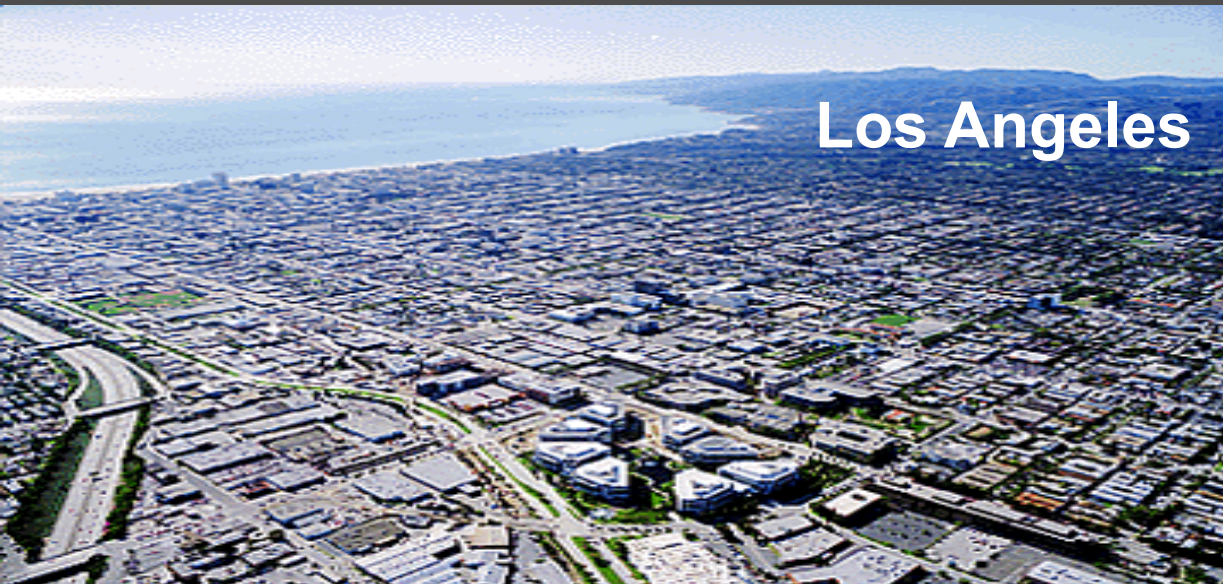
Constrained location

SF (CA), LA (CA), London

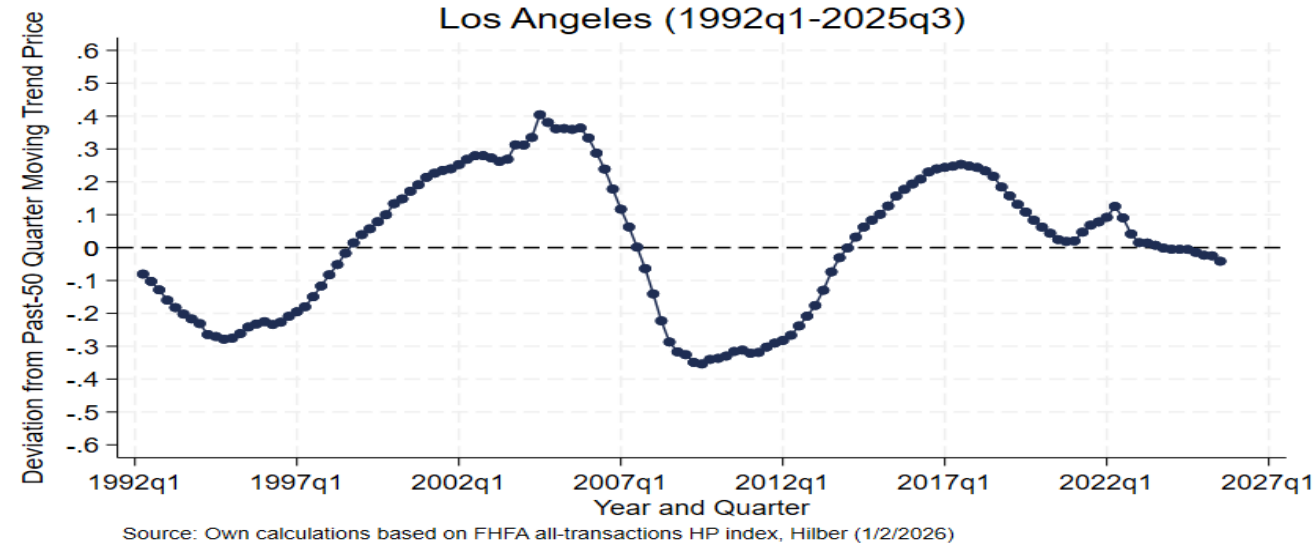


⇒ Housing stock is durable!

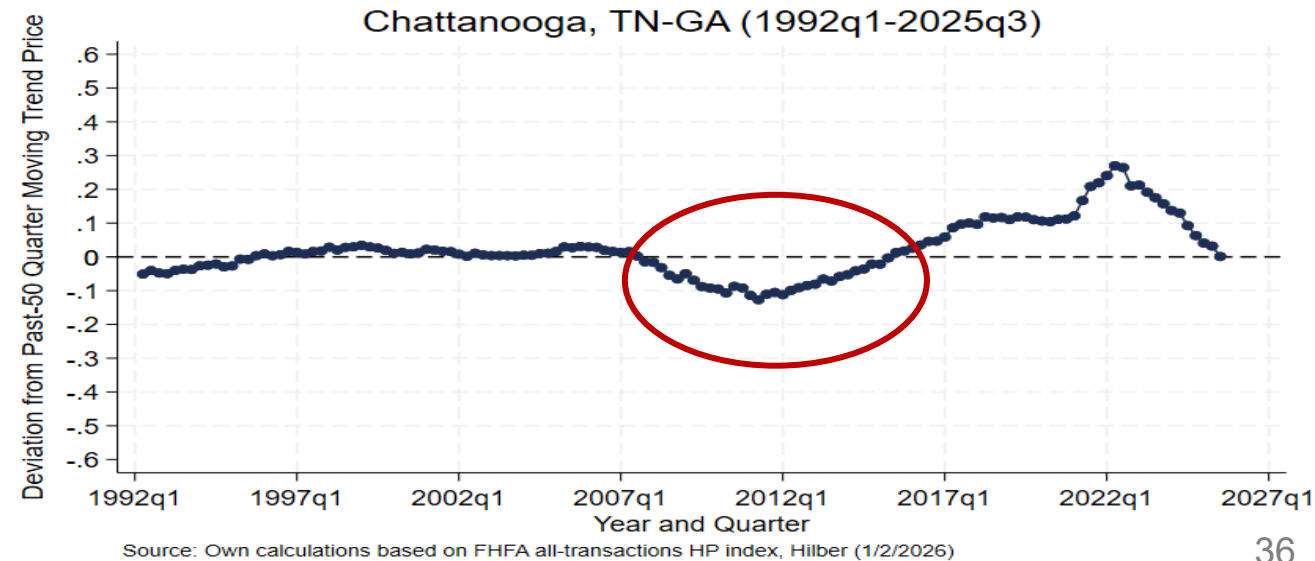
Other examples...



Los Angeles

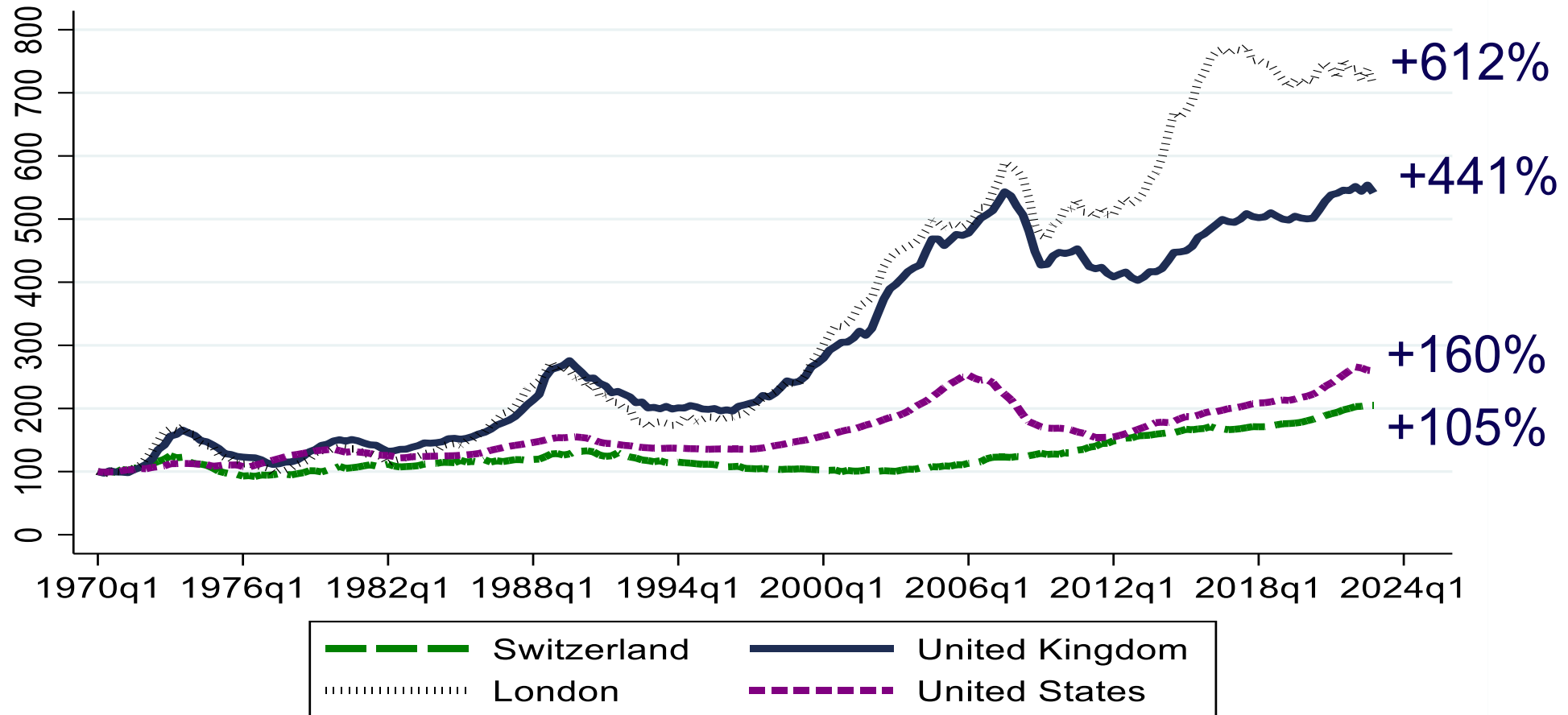


Chattanooga, TN-GA



And at country level...

Real house prices: Switzerland vs. US vs. UK (& London)
(Price indices; 1970q1=100; 1970q1-2021q4)



Source 1: National sources, BIS Residential Property Price database (<http://www.bis.org/statistics/pp.ht>)

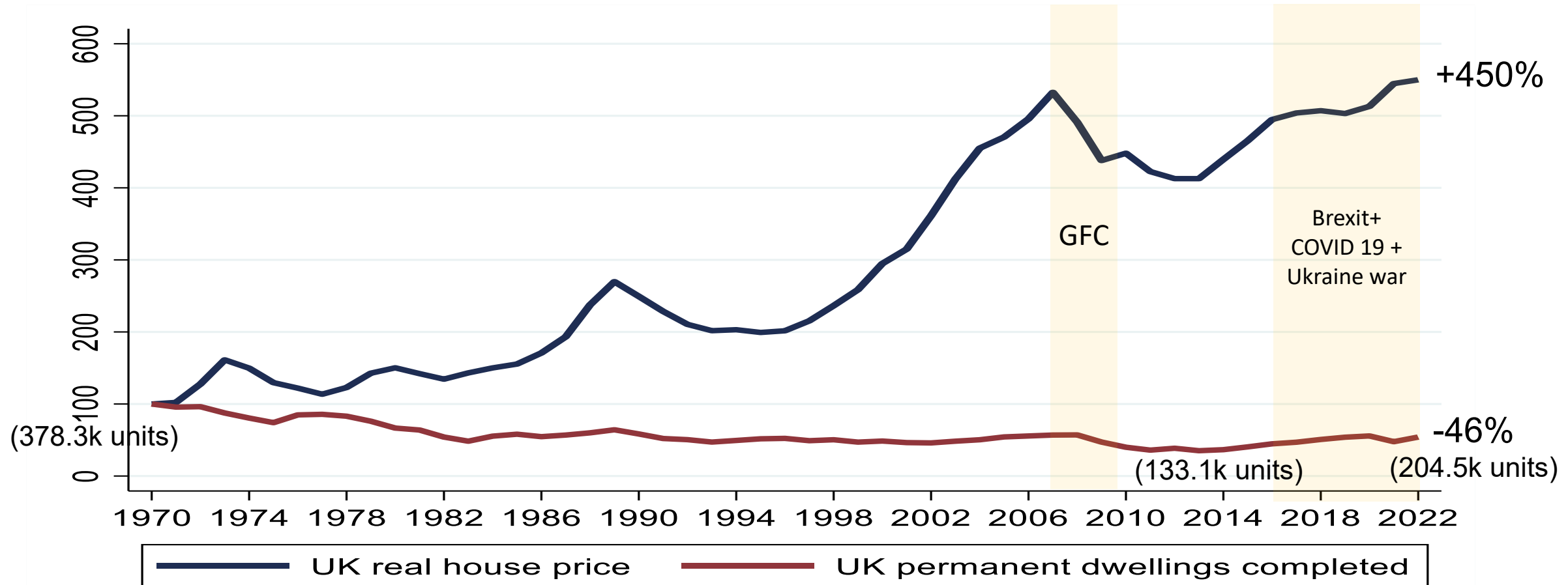
Source 2: For London: Land Registry (nominal HPI) & FRED (CPI)

Notes: Real house price indices, 1970q1=100, using CPI to deflate nominal HPI, Hilber (2023)

Why these differences b/w UK, US & Switzerland? (Discuss)

What about corresponding construction numbers?

UK real house price and construction indices (1970-2022)



Sources: BIS Residential Property Price Database & Office for National Statistics

Weblinks: <http://www.bis.org/statistics/pp.htm> & [ons.gov.uk \[housing\]](https://ons.gov.uk/housing)

Notes: Real HPI (yearly av.) & construction idx, 1970=100), using CPI to deflate nom. HPI, Hilber (2023)

What about corresponding construction numbers? (Cont.)

UK vs. US vs. Switzerland

	Newly constructed units (in 2010)	Population in Million (in 2010)	Ratio in % (in 2010)	Ratio in % (in 2015)
UK	137,260	62.77	0.22%	0.21%
US	866,000	309.3	0.28%	0.47%
CH	50,166	7.825	0.66%	NA

Source: OECD Housing Affordability Database, Population Censuses, own calculations

⇒ Switzerland builds about 3 x as many units p.c. as UK (+ units are perhaps 50% larger)!

Some first tentative conclusions

1. **Real estate cycles are ‘local’** in nature, and **magnitude of cycles varies** a lot across markets
2. **Long-run supply price elasticity matters in practice** because market demand curve is downward sloping
3. **Supply curve is kinked**, suggesting asymmetric response to positive and negative D shocks

Preparations for **Seminar 4** (Topic 4, Week 8)

- **Small group presentations with discussion**
 - ▶ **Topic:** Housing market dynamics around the globe – facts and determinants
 - ▶ **Details:** See special handout (on Moodle)
 - ▶ **Aim:** Illustrate housing market dynamics in different markets and try to explain...can we make sense of empirical facts?

Preparations for **Seminar 5** (Topics 4/5/6, Week 9)

- **Topic:** Property Market Dynamics: Theory and Empirical Tests
- **Seminar format:** Group presentations of answer to past exam question & subsequent discussion
- **Key readings** for seminar:
 - a. Compulsory readings for Topics 4/5/6 (pick two theories/papers; e.g., Case, Shiller & Thompson, 2012 & Hilber and Vermeulen, 2016)
 - b. Capozza *et al.* (2004)
 - c. DeFusco *et al.* (2018), Nathanson & Zwick (2018)
- **Details:** See separate handout (on Moodle)